Quadratic equations and parabolas

1

a

i Maximum ii Negative iii 2 solutions

b

i Minimum ii Positive iii 0 solutions

c

i Maximum ii Negative iii 1 solution

d

i Minimum ii Positive iii 2 solutions

2 The equation has no real solution.

3

a 2 solutions b 1 solution c 0 solutions

The discriminant and equations

4

a

i $\Delta = 0$ ii 1

b

i $\Delta = -76$ ii 0

c

i $\Delta = 1$ ii 2

d

i $\Delta = 16$ ii 2

e

i $\Delta = 0$ ii 1

f

i $\Delta = -23$ ii 0

g

i $\Delta = 40$ ii 2

h

i $\Delta = 100$ ii 2

5

a $\Delta = 0$ b Rational

6

a

i $\Delta = -144$ ii Non-real roots.

b

i $\Delta = 112$ ii Real, unequal and irrational roots.

c

i $\Delta = 0$ ii Real, equal and rational roots.

d

i $\Delta = 64$ ii Real, unequal and rational roots.

e

i $\Delta = 0$ ii Real, equal and rational roots.

f

i $\Delta = 65$ ii Real, unequal and irrational roots.

g

i $\Delta = -4$ ii Non-real roots.

h

i $\Delta = 3.75$ ii Real, unequal and irrational roots.

7

a $\Delta = 256$ b The roots are unequal and rational.

c $x = 12, x = -4$

8 There is only one real solution.

9 $b^2 - 4ac > 0$

10

a $9 + 8m$ b $16n^2 + 8$ c $45 - 4p$ d $16 + 4p^2$

e $12m$ f $\frac{-23n^2}{4}$

11 $a < \frac{9}{10}$

12 $m > -1$

13

a $m > -\frac{9}{20}$

b $m = 0$ must be eliminated as m is the coefficient of x^2 and by definition a quadratic equation must have an x^2 term.

14 $n = \pm 9$

15 $k = 1$

16 $m = -2, m = 4$

17

a

i $64-8k$ ii $k=8$ iii $k \leq 8$ iv $k>8$

v $k<8$

b

i $400-8k$ ii $k=50$ iii $k \leq 50$ iv $k>50$

v $k<50$

c

i $52-12k$ ii $k=\frac{13}{3}$ iii $k \leq \frac{13}{3}$ iv $k>\frac{13}{3}$

v $k<\frac{13}{3}$

18

a $k>74$ b $k=75$

The discriminant and parabolas

19 None

20 One x-intercept

21

a i $a>0$ ii $\Delta>0$

b i $a<0$ ii $\Delta=0$

c i $a<0$ ii $\Delta<0$

d i $a>0$ ii $\Delta<0$

e i $a>0$ ii $\Delta<0$

f i $a>0$ ii $\Delta>0$

g i $a<0$ ii $\Delta=0$

h i $a<0$ ii $\Delta>0$

i i $a>0$ ii $\Delta=0$

j i $a<0$ ii $\Delta<0$

k i $a<0$ ii $\Delta=0$

l i $a<0$ ii $\Delta>0$

m i $a<0$ ii $\Delta<0$

n i $a>0$ ii $\Delta=0$

o i $a<0$ ii $\Delta>0$

p i $a>0$ ii $\Delta<0$

22 **a** $m < -\frac{27}{4}$ **b** $m = -7$

23 $k = 16$

Application

24

a $0 < p < q$

b $x^2 + 2(p - q)x + p^2 - q^2 = 0$

c $\Delta = 8q(q - p)$

d Two real solutions as $q > p$, therefore the discriminant is positive.

e $x = 3p$